10.2.1 A simple QR algorithm

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Week 10 due Mar 27, 2025 22:10 IST

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it also says start with a matrix of V equal to I. But now in each iteration

take your matrix A, compute its QR factorization, and once you have its QR

factorization, multiply R times Q, creating a new

matrix A. This is no longer the same matrix as we started with. And then take

this matrix Q and multiply it from the right times our original matrix V, or the V

that we're now accumulating. And

4:08 / 4:08

2.0x

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Reading assignment

1/1 point (graded)

Read Unit 10.2.1 of the notes. [LINK]

- [x] done

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Homework 10.2.1.1

1/1 point (graded)
Show that for the algorithm on the right
$A^{(k+1)} = Q^{(k+1)\,H} A^{(k)} Q^{(k+1)}$.

- [x] done

Solution

For a complete solution, see the notes .

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Answers are displayed within the problem

Homework 10.2.1.2

1/1 point (graded)
In the above algorithms, for all k,

- $\widehat{A}^{(k)} = A^{(k)}$.
- $\widehat{R}^{(k)} = R^{(k)}$.
- $\widehat{V}^{(k)} = V^{(k)}$.

- [x] done

Solution

For a complete solution, see the notes .

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Homework 10.2.1.3

1/1 point (graded)
In the above algorithms, show that for all k

- $V^{(k)} = Q^{(0)} Q^{(1)} \cdots Q^{(k)}$.
- $A^k = V^{(k)} R^{(k)} \cdots R^{(1)} R^{(0)}$. (Note: A^k here denotes A raised to the kth power.)

- [x] done

Solution

For a complete solution, see the notes .

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becomes the diagonal matrix. And if you

accumulate all of these transformations of the basis that you encounter along

the way, what you end up with is a matrix where the columns of that matrix

equal the eigenvectors associated with the eigenvalues that appear on the

diagonal of matrix A. And that's how the two are related to each other. But this

is surprisingly simple. Compute the QR

3:53 / 3:53

2.0x

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Homework 10.2.1.4

1/1 point (graded)

Examine Assignments/Week10/matlab/SubspaceIterationAllVectors.m, which implements the subspace iteration in Figure 10.2.1.1 (left). Examine it by executing the script in Assignments/Week10/matlab/test_simple_QR_algorithms.m.

- [x] done

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Homework 10.2.1.5

1/1 point (graded)

Copy Assignments/Week10/matlab/SubspaceIterationAllVectors.m into `SimpleQRAlg.m` and modify it to implement the algorithm in Figure 10.2.1.1 (right) as

```
function [ Ak, V ] = SimpleQRAlg( A, maxits, illustrate, delay )
```

Modify the appropriate line in Assignments/Week10/matlab/test_simple_QR_algorithms.m, changing `(0)` to `(1)`, and use it to examine the convergence of the method.

- [x] done

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